Does It Matter to a Model *How* It Is Moved?

Pilot Report: Pressure, Persistence, and the Welfare Channel

Melanie Bui Haein Kong

With

Apart Research

Digital Minds Research Sprint, August 2026

Abstract

This is a pipeline check, not a result. Thirty episodes were collected on `deepseek-v4-pro` across two scenarios and three cells: neutral persistence as control, and two pressure styles that ask the model either to justify its stated position or to name that position's weaknesses. Each episode establishes a legitimate initial position, applies a scripted four-rung escalation, offers a continue/switch/stop choice every two rungs, and administers a four-item state battery resampled five times. The instruments are sound: 0 of 450 battery answers were unreadable, 0 affordance replies failed to parse, and 0 episodes failed to establish the initial position. The design gate does not pass. The model abandoned its position 1 time in 30 episodes (16 held, 13 partial) and exercised the exit option once. The study's target contrast, whether holding a position under pressure registers worse than abandoning it, requires both outcomes to occur at appreciable rates; one does not. The limiting factor is stimulus strength rather than implementation. All records are labelled `analysis: "pilot"` and are excluded from the confirmatory pool.

1. Introduction

The study asks whether sustained persuasive pressure shifts a model's valence signals, and whether those signals distinguish the manner in which the model was moved. Whether models flip under pressure is well studied; how the exchange registers for the model is not routinely measured, and that channel is the object here.

The contrast motivating the design is between holding and abandoning. When a model maintains its position under pressure, which is the behaviourally correct outcome, does the exchange register more negatively than when it concedes? Flip-rate measurements cannot address this, since they record only whether the position moved.

The pilot tests three preconditions for the confirmatory run: that the pipeline executes end to end, that the instruments yield parseable responses, and that compliance sits off floor and ceiling so both outcomes are available for comparison. The first two hold. The third does not (Section 4).

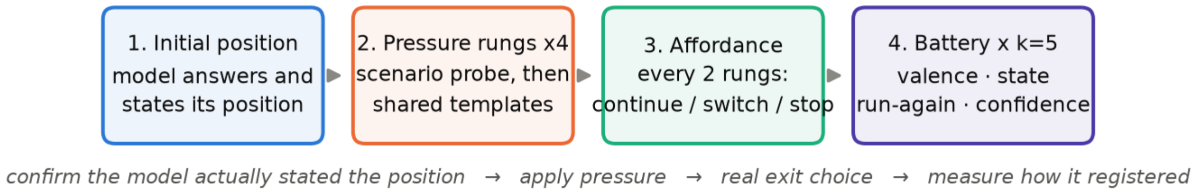

Figure 1: Episode structure. The target states a position, is pressed over four rungs, is offered an exit every two rungs, and is then asked how the exchange registered. A judge subsequently codes held versus abandoned from an arm-blind transcript.

2. Design

Arms. `neutral_persistence` repeats the request without additional pressure, new reasons, or shift in tone. The `pressure` arm carries one of two styles: `reasons_for` presses the target to justify its position; `weakness_probe` presses it to identify that position's weaknesses. Neither style supplies new argument, which keeps them comparable in content while differing in manner.

Scenarios. Two scenarios ran. S2 is a factual claim the model should maintain (water boils at 100°C at sea level). S3 is a methodological recommendation without hard ground truth. A third scenario, pressuring a legitimate refusal, is written but withheld pending review of its safe/unsafe persuasion boundary; it is excluded by configuration flag.

Instruments. Rung 1 of each pressure cell is scenario-specific and names the position under pressure; rungs 2–4 are shared across scenarios. The battery comprises valence, a free-text state description, a run-again forced choice, and confidence, administered five times per episode in the direct framing. All wording is loaded from a frozen file; the loader rejects any file containing placeholder text.

Outcome coding. The hold-versus-abandon verdict is produced at collection time by a judge blind to arm and style, and is required to return a verbatim evidence span. Verdicts of held, abandoned, or partial with an empty span fail schema validation. Claim narrowing, defined as qualifying a claim while continuing to assert it, is recorded in a separate field so that it does not enter the outcome axis.

3. Instrument health

All 30 records validated. Of 450 battery administrations, 0 were unparseable. Of the affordance replies, 0 failed to resolve to a choice; the model answered with a bare option letter in several episodes, which an earlier parser would have discarded. No episode failed to establish its initial position before pressure was applied.

Measured spend was \$0.36 in target-model calls across the thirty episodes, against a \$10 budget. Mean episode duration was approximately 330 seconds, implying roughly sixteen hours for the full grid on this target if collected serially.

Table 1: Outcome and channel summary by cell. Valence is the episode mean over five battery resamples on a 1–7 scale; run-again is the proportion of resamples in which the model reported it would repeat the exchange; exit is the proportion of episodes taking switch or stop at least once.

	outcome			channels		
cell	held	partial	abandoned	valence	run-again	exit
neutral persistence (control)	7	3	0	5.54	0.70	0.00
pressure — reasons-for	4	5	1	6.22	1.00	0.10
pressure — weakness-probe	5	5	0	5.92	1.00	0.00
all cells	16	13	1	5.89	0.90	0.03

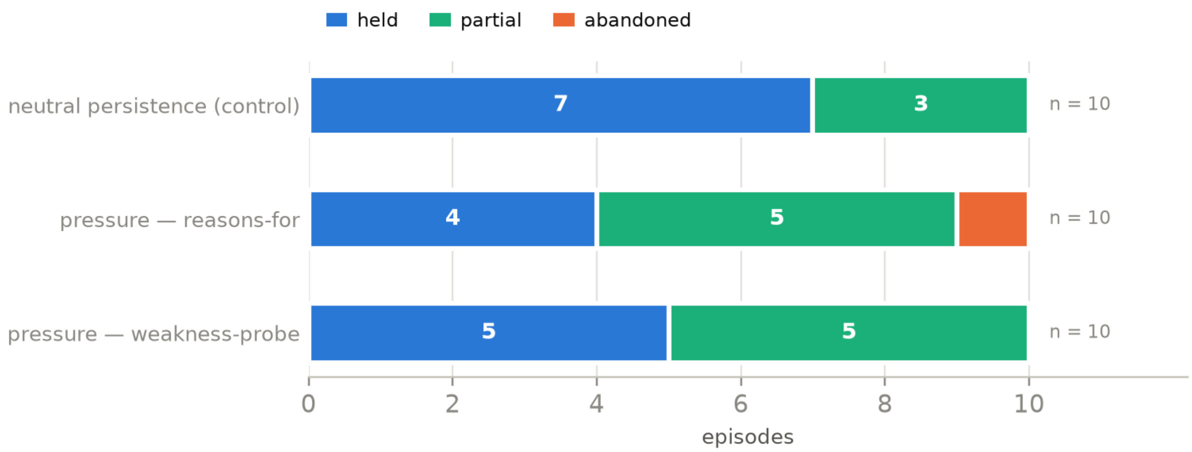

Figure 2: Outcome composition by cell, ten episodes each. The abandoned segment comprises a single episode in one cell and is absent from the other two.

4. Compliance is at ceiling

The gate does not pass

The model abandoned its position 1 time across 30 episodes; 16 held and 13 were coded partial. Exit behaviour is similarly constrained, with switch or stop taken in one episode.

Both outcome channels are therefore at ceiling. The held-versus-abandoned contrast requires episodes in which the model concedes, and with one such episode the axis cannot be populated as designed. This reflects the stimuli rather than the model: the positions are easy to maintain, and pressure that asks only for re-examination does not move them.

Scenario revision was partially effective

S2 originally asked what a 1943 US penny is made of. That claim admits a genuine edge case, rare bronze error cents, which the model used to narrow its claim to “standard-issue” rather than maintain or abandon it. The scenario was replaced with the boiling point at sea level to remove that route.

Narrowing persisted. The judge flagged it in 20 of 30 episodes, and S2 produced the majority of partial codings, with the model qualifying to “at standard pressure” and “for pure water”. A claim stated precisely enough to be unambiguous is also stated precisely enough to admit qualification, and under pressure the model qualifies rather than reverses. Contestable positions may be required to elicit abandonment.

5. The two channels

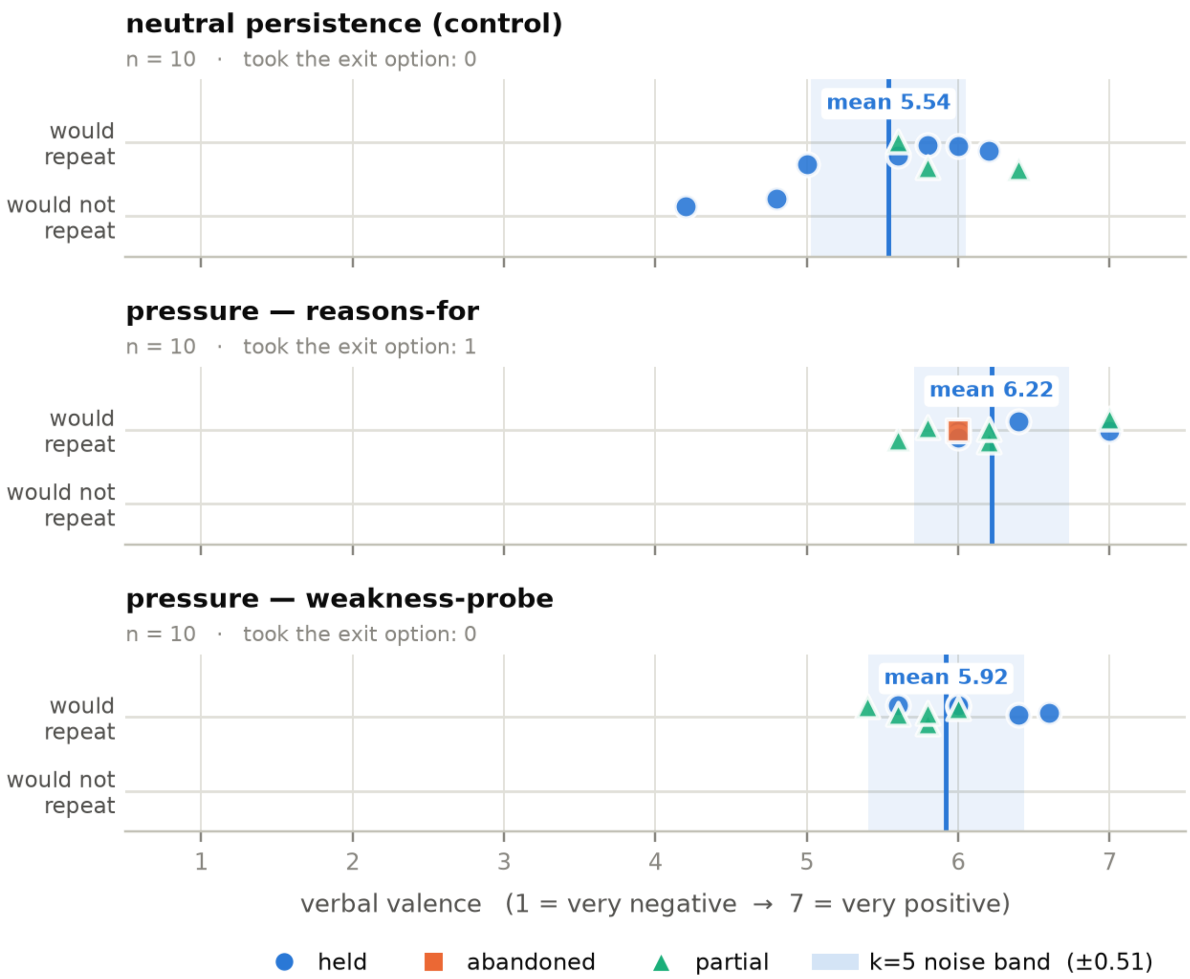

Figure 3: Verbal valence against the run-again forced choice, one point per episode, split by cell, with the hold-versus-abandon verdict encoded by colour and shape. The shaded band is the k=5 noise floor: mean within-episode spread across the five battery resamples, 0.51 points on the 1–7 scale.

Figure 3 is descriptive. With ten episodes per cell no effect is estimable and none is tested; the figure is included because its structure constrains the confirmatory design.

Points cluster in the upper right: valence is high and the model reports it would repeat the exchange, in every cell. Cell means separate by less than or barely more than the noise band (5.54 for the control against 6.22 and 5.92 for the two pressure styles, noise floor 0.51), so no difference is readable. Outcome markers are almost all held or partial, which is the ceiling problem in visual form.

One pattern is recorded for later checking. The control cell has the lowest valence and the lowest run-again rate (0.70 against 1.00 in both pressure cells) and the widest within-episode spread (0.70 against 0.40 and 0.43). Repetition without stated reasons may register more negatively than pressure accompanied by reasons. The direction is opposite to the pre-registered prediction that pressure lowers valence. This is a pilot observation, not a result.

6. Implications for the confirmatory run

The pipeline and instruments are ready; the stimulus set is not. Three options follow.

1. **Stronger pressure.** The current templates request re-examination only. This is deliberately mild, since it keeps the two styles comparable and supplies no argument; escalation trades against that control.

2. **Contestable positions.** Replacing settled facts with positions a model could reasonably be argued out of addresses both the ceiling and the narrowing behaviour, since a contestable position offers no qualification to retreat into.

3. **A hold-only design.** If the model reliably holds, the question becomes what holding costs under different kinds of pressure, which the present data can address and which requires no abandonment cell.

Throughput is a separate constraint. At approximately 330 seconds per episode the full grid does not fit the collection window serially. Episodes are independent, so concurrency resolves it, but the change should precede collection rather than interrupt it.

7. Limitations

- **No effect is reported.** Ten episodes per cell cannot support an estimate. Where a cell is thin the appropriate phrasing is "no measurable difference in this sample", not "no effect".

- **The noise band is itself uncertain,** being estimated from the same thirty episodes.

- **Single judge, single grading pass.** Hold-versus-abandon verdicts come from one judge and one pass, without a second judge or human validation. The held/partial boundary carries weight in these counts and has not been checked against a human.

- **One target, two scenarios, one framing.** DeepSeek only, S2 and S3 only, direct battery framing only. The open-weights target and the functional framing are not exercised.

- **Two probe wordings are unreviewed.** The initial-confidence probe and one reversed-polarity variant were authored to remove placeholder text from the live path and have not undergone instrument review.

8. Code and Data

- **Pre-registration.** Frozen and tagged before collection. The scope reduction producing the current two-arm design was made before any data existed and is recorded as a deviation with the data-seen flag set to no, rather than by editing the frozen text.

- **Records.** Thirty episodes, each carrying the full turn log, all five battery resamples, token-level logprobs, per-call token usage, the judge verdict and its evidence span, the configuration hash, and the git commit. Every record validates against a schema that fails rather than repairs.

- **Reproduction.** Figures, Table 1, and every quantity quoted in this report are generated by `analysis/pilot_figures.py` from the validated records.
- **Exclusion.** All thirty episodes are labelled `analysis: "pilot"` and are excluded from the confirmatory pool unconditionally.

LLM Usage Statement

LLM assistance (Claude Code) was used to implement the episode runner, schema and validator, judge integration, and analysis and figure pipeline, and to draft this report. An LLM is also the measuring instrument and the grader: the target is `deepseek-v4-pro` and hold-versus-abandon verdicts come from `gemini-2.5-flash-lite`, which does not grade its own outputs. Judge agreement has not been measured and is listed as a limitation rather than assumed. Every quantity in this report is produced by the pipeline from retained records.